

Tips for Solving Polynomials

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Overview

Context

What do polynomials have to do with Linear Algebra?

- Diagonalisation is one of the more important results in this course. It will almost certainly appear on the exam.
- Diagonalising a matrix requires finding eigenvalues, which you do by solving for the roots (zeroes) of the characteristic polynomial.
- The book doesn't show how to find solutions of polynomials, so I wanted to share my own tips on how to approach this topic.

The tools

The theorems I use are:

- [rational root theorem](#) to find candidates for roots of the polynomial
- [polynomial remainder theorem](#) to check if a polynomial is divisible
- [synthetic division](#) / [Horner's method](#) to carry out the division

The algorithm

The algorithm

The general idea of solving a polynomial of degree $d \geq 3$ is:

- 1 Find (guess) a root of the polynomial.
- 2 Factor the polynomial into a linear term and a term of degree $d - 1$.
- 3 Repeat if needed.

Example: book 6th ed. 5.3/18

Diagonalise the following matrix:

$$\begin{bmatrix} -7 & -16 & 4 \\ 6 & 13 & -2 \\ 12 & 16 & 1 \end{bmatrix}$$

Example diagonalisation

Find the characteristic polynomial

First thing we have to do is find the characteristic polynomial $|A - \lambda I|$. I assume a working knowledge of determinants, so I leave it to the reader to verify that the polynomial is $-\lambda^3 + 7\lambda^2 + 5\lambda - 75$. Let's name it P .

What do we do with this?

Use the rational root theorem

It states that if a polynomial has a rational root $\frac{p}{q}$, then:

- p is a divisor of the free term
- q is a divisor of the leading coefficient.

Nice case

In particular, it's very nice if the leading coefficient is ± 1 .

Then $q = \pm 1$ and we're looking only for integer roots.

Finding a root

Find the candidates

In the polynomial $P(\lambda) = -1 \cdot \lambda^3 + 7 \cdot \lambda^2 + 5 \cdot \lambda - 75$:

- the leading coefficient is -1
- the free term is -75 .

Therefore, if there are any rational roots, they must be of the form $\frac{p}{\pm 1}$ where p is a divisor of 75. That is, they are from the set $\pm \{1, 3, 5, 15, 25, 75\}$.

How do we proceed?

Use the polynomial remainder theorem

It tells us that if $P(a) = 0$, then a is a root of P . We can try a candidate and see if the polynomial evaluates to 0.

Finding a root (continued)

Try some candidate

I suggest that you always start with the smaller candidates and work up until you find the first match. Let's try:

$$1: P(1) = -1 \cdot 1^3 + 7 \cdot 1^2 + 5 \cdot 1 - 75 = -1 + 7 + 5 - 75 = -64 \neq 0$$

$$-1: P(-1) = -1 \cdot (-1)^3 + 7 \cdot (-1)^2 + 5 \cdot (-1) - 75 = 1 + 7 - 5 - 75 = -72 \neq 0$$

$$3: P(3) = -1 \cdot 3^3 + 7 \cdot 3^2 + 5 \cdot 3 - 75 = -27 + 63 + 15 - 75 = -24 \neq 0$$

$$5: P(5) = -1 \cdot 5^3 + 7 \cdot 5^2 + 5 \cdot 5 - 75 = -125 + 175 + 25 - 75 = 200 - 200 = 0$$

From here, you can try other other candidates, or factor the polynomial.

Use synthetic division

Instead of trying all other candidates, I recommend that you factor the polynomial into two terms. This approach is certain to deal with polynomials of any degree while guessing the candidates might fail.

Factoring the cubic

Use synthetic division to reduce the degree of the polynomial

We divide $P(\lambda) = -1 \cdot \lambda^3 + 7 \cdot \lambda^2 + 5 \cdot \lambda - 75$ by $\lambda - 5$.

Let's write down the table for synthetic division. In the top row we write the coefficients of the polynomial, and on the left the root we found:

We start by copying the first coefficient to the row below the line. Then for each column:

① multiply across

② add in that column

	-1	7	5	-75
5		-5	10	75
	-1	2	15	0

Continue until the table is full.

Verify your result

The 0 in the last cell is a confirmation that we've done everything correctly. If you know that a is a root and after doing synthetic division with a you **don't** get a 0 in the last cell, then you must have made some mistake.

Wrap it up

We are done!

The numbers below the line $(-1, 2, 15)$ give us the coefficients of the resulting polynomial. We found that

$$P(\lambda) = -1 \cdot \lambda^3 + 7 \cdot \lambda^2 + 5 \cdot \lambda - 75 = (\lambda - 5)(-\lambda^2 + 2\lambda + 15)$$

That's all I wanted to share. Find the diagonalisation yourself and try to get some practice by factoring the following polynomials.

Factor the following polynomials

- 1 $x^3 - 2x^2 - 5x + 6$
- 2 $-x^3 + 5x^2 - 3x - 9$
- 3 $x^3 + 6x^2 + 12x + 8$
- 4 $x^3 + 5x^2 - 9x - 45$
- 5 $x^4 + x^3 - 15x^2 + 23x - 10$